$3d^7$

August 25, 2026

1 $\left|3d_{xz}^2 3d_{z^2}^2 3d_{x^2-y^2}^2 3d_{xy}^\alpha 3d_{yz}^0\right\rangle$

Here are the CSFs:

$$|1,3/2,3/2\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha\right\rangle$$
$$|2,3/2,3/2\rangle = \left|3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha\right\rangle$$
$$|3,3/2,3/2\rangle = \left|3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\alpha\right\rangle$$
$$|4,3/2,3/2\rangle = \left|3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^2\right\rangle$$
$$|5,3/2,3/2\rangle = \left|3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha\right\rangle$$
$$|6,3/2,3/2\rangle = \left|3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\alpha\right\rangle$$
$$|7,3/2,3/2\rangle = \left|3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^2\right\rangle$$
$$|8,3/2,3/2\rangle = \left|3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha\right\rangle$$
$$|9,3/2,3/2\rangle = \left|3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2\right\rangle$$
$$|10,3/2,3/2\rangle = \left|3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$
$$|11,3/2,1/2\rangle = \frac{1}{\sqrt{3}}\left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha\right\rangle$$

$$
\begin{aligned}
|12, 3/2, 1/2\rangle =& \frac{1}{\sqrt{3}} \left| 3d_{xz}^{2} 3d_{yz}^{\alpha} 3d_{xy}^{2} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{\beta} \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^{2} 3d_{yz}^{\alpha} 3d_{xy}^{2} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\alpha} \right\rangle \\
&+ \frac{1}{\sqrt{3}} \left| 3d_{xz}^{2} 3d_{yz}^{\beta} 3d_{xy}^{2} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{\alpha} \right\rangle \\
|13, 3/2, 1/2\rangle =& \frac{1}{\sqrt{3}} \left| 3d_{xz}^{2} 3d_{yz}^{\alpha} 3d_{xy}^{\alpha} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\beta} \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^{2} 3d_{yz}^{\alpha} 3d_{xy}^{\beta} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\alpha} \right\rangle \\
&+ \frac{1}{\sqrt{3}} \left| 3d_{xz}^{2} 3d_{yz}^{\beta} 3d_{xy}^{\alpha} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\alpha} \right\rangle \\
|14, 3/2, 1/2\rangle =& \frac{1}{\sqrt{3}} \left| 3d_{xz}^{2} 3d_{yz}^{\alpha} 3d_{xy}^{\alpha} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{2} \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^{2} 3d_{yz}^{\alpha} 3d_{xy}^{\beta} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{2} \right\rangle \\
&+ \frac{1}{\sqrt{3}} \left| 3d_{xz}^{2} 3d_{yz}^{\beta} 3d_{xy}^{\alpha} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{2} \right\rangle \\
|15, 3/2, 1/2\rangle =& \frac{1}{\sqrt{3}} \left| 3d_{xz}^{\alpha} 3d_{yz}^{2} 3d_{xy}^{2} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{\beta} \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^{\alpha} 3d_{yz}^{2} 3d_{xy}^{2} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\alpha} \right\rangle \\
&+ \frac{1}{\sqrt{3}} \left| 3d_{xz}^{\beta} 3d_{yz}^{2} 3d_{xy}^{2} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{\alpha} \right\rangle \\
|16, 3/2, 1/2\rangle =& \frac{1}{\sqrt{3}} \left| 3d_{xz}^{\alpha} 3d_{yz}^{2} 3d_{xy}^{\alpha} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\beta} \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^{\alpha} 3d_{yz}^{2} 3d_{xy}^{\beta} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\alpha} \right\rangle \\
&+ \frac{1}{\sqrt{3}} \left| 3d_{xz}^{\beta} 3d_{yz}^{2} 3d_{xy}^{\alpha} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\alpha} \right\rangle \\
|17, 3/2, 1/2\rangle =& \frac{1}{\sqrt{3}} \left| 3d_{xz}^{\alpha} 3d_{yz}^{2} 3d_{xy}^{\alpha} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{2} \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^{\alpha} 3d_{yz}^{2} 3d_{xy}^{\beta} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{2} \right\rangle \\
&+ \frac{1}{\sqrt{3}} \left| 3d_{xz}^{\beta} 3d_{yz}^{2} 3d_{xy}^{\alpha} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{2} \right\rangle \\
|18, 3/2, 1/2\rangle =& \frac{1}{\sqrt{3}} \left| 3d_{xz}^{\alpha} 3d_{yz}^{\alpha} 3d_{xy}^{2} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\beta} \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^{\alpha} 3d_{yz}^{\beta} 3d_{xy}^{2} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\alpha} \right\rangle \\
&+ \frac{1}{\sqrt{3}} \left| 3d_{xz}^{\beta} 3d_{yz}^{\alpha} 3d_{xy}^{2} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\alpha} \right\rangle \\
|19, 3/2, 1/2\rangle =& \frac{1}{\sqrt{3}} \left| 3d_{xz}^{\alpha} 3d_{yz}^{\alpha} 3d_{xy}^{2} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{2} \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^{\alpha} 3d_{yz}^{\beta} 3d_{xy}^{2} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{2} \right\rangle \\
&+ \frac{1}{\sqrt{3}} \left| 3d_{xz}^{\beta} 3d_{yz}^{\alpha} 3d_{xy}^{2} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{2} \right\rangle \\
|20, 3/2, 1/2\rangle =& \frac{1}{\sqrt{3}} \left| 3d_{xz}^{\alpha} 3d_{yz}^{\alpha} 3d_{xy}^{\beta} 3d_{z^2}^{2} 3d_{x^2-y^2}^{2} \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^{\alpha} 3d_{yz}^{\beta} 3d_{xy}^{\alpha} 3d_{z^2}^{2} 3d_{x^2-y^2}^{2} \right\rangle \\
&+ \frac{1}{\sqrt{3}} \left| 3d_{xz}^{\beta} 3d_{yz}^{\alpha} 3d_{xy}^{\alpha} 3d_{z^2}^{2} 3d_{x^2-y^2}^{2} \right\rangle
\end{aligned}
$$

$$\begin{aligned}
|21,3/2,-1/2\rangle =& \frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|22,3/2,-1/2\rangle =& \frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|23,3/2,-1/2\rangle =& \frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|24,3/2,-1/2\rangle =& \frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle \\
&+ \frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle \\
|25,3/2,-1/2\rangle =& \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|26,3/2,-1/2\rangle =& \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|27,3/2,-1/2\rangle =& \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle \\
&+ \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle \\
|28,3/2,-1/2\rangle =& \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|29,3/2,-1/2\rangle =& \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle \\
&+ \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle
\end{aligned}$$

$$|30,3/2,-1/2\rangle = \frac{1}{\sqrt{3}}\left|3d^{\alpha}_{xz}3d^{\beta}_{yz}3d^{\beta}_{xy}3d^{2}_{z^2}3d^{2}_{x^2-y^2}\right\rangle + \frac{1}{\sqrt{3}}\left|3d^{\beta}_{xz}3d^{\alpha}_{yz}3d^{\beta}_{xy}3d^{2}_{z^2}3d^{2}_{x^2-y^2}\right\rangle + \frac{1}{\sqrt{3}}\left|3d^{\beta}_{xz}3d^{\beta}_{yz}3d^{\alpha}_{xy}3d^{2}_{z^2}3d^{2}_{x^2-y^2}\right\rangle$$

$$|31,3/2,-3/2\rangle = \left|3d^{2}_{xz}3d^{2}_{yz}3d^{\beta}_{xy}3d^{\beta}_{z^2}3d^{\beta}_{x^2-y^2}\right\rangle$$

$$|32,3/2,-3/2\rangle = \left|3d^{2}_{xz}3d^{\beta}_{yz}3d^{2}_{xy}3d^{\beta}_{z^2}3d^{\beta}_{x^2-y^2}\right\rangle$$

$$|33,3/2,-3/2\rangle = \left|3d^{2}_{xz}3d^{\beta}_{yz}3d^{\beta}_{xy}3d^{2}_{z^2}3d^{\beta}_{x^2-y^2}\right\rangle$$

$$|34,3/2,-3/2\rangle = \left|3d^{2}_{xz}3d^{\beta}_{yz}3d^{\beta}_{xy}3d^{\beta}_{z^2}3d^{2}_{x^2-y^2}\right\rangle$$

$$|35,3/2,-3/2\rangle = \left|3d^{\beta}_{xz}3d^{2}_{yz}3d^{2}_{xy}3d^{\beta}_{z^2}3d^{\beta}_{x^2-y^2}\right\rangle$$

$$|36,3/2,-3/2\rangle = \left|3d^{\beta}_{xz}3d^{2}_{yz}3d^{\beta}_{xy}3d^{2}_{z^2}3d^{\beta}_{x^2-y^2}\right\rangle$$

$$|37,3/2,-3/2\rangle = \left|3d^{\beta}_{xz}3d^{2}_{yz}3d^{\beta}_{xy}3d^{\beta}_{z^2}3d^{2}_{x^2-y^2}\right\rangle$$

$$|38,3/2,-3/2\rangle = \left|3d^{\beta}_{xz}3d^{\beta}_{yz}3d^{2}_{xy}3d^{2}_{z^2}3d^{\beta}_{x^2-y^2}\right\rangle$$

$$|39,3/2,-3/2\rangle = \left|3d^{\beta}_{xz}3d^{\beta}_{yz}3d^{2}_{xy}3d^{\beta}_{z^2}3d^{2}_{x^2-y^2}\right\rangle$$

$$|40,3/2,-3/2\rangle = \left|3d^{\beta}_{xz}3d^{\beta}_{yz}3d^{\beta}_{xy}3d^{2}_{z^2}3d^{2}_{x^2-y^2}\right\rangle$$

$$|41,1/2,1/2\rangle = +\frac{1}{\sqrt{2}}\left|3d^{2}_{xz}3d^{2}_{yz}3d^{\alpha}_{xy}3d^{\beta}_{z^2}3d^{\alpha}_{x^2-y^2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d^{2}_{xz}3d^{2}_{yz}3d^{\beta}_{xy}3d^{\alpha}_{z^2}3d^{\alpha}_{x^2-y^2}\right\rangle$$

$$|42,1/2,1/2\rangle = +\frac{1}{\sqrt{2}}\left|3d^{2}_{xz}3d^{\alpha}_{yz}3d^{2}_{xy}3d^{\beta}_{z^2}3d^{\alpha}_{x^2-y^2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d^{2}_{xz}3d^{\beta}_{yz}3d^{2}_{xy}3d^{\alpha}_{z^2}3d^{\alpha}_{x^2-y^2}\right\rangle$$

$$|43,1/2,1/2\rangle = +\frac{1}{\sqrt{2}}\left|3d^{2}_{xz}3d^{\alpha}_{yz}3d^{\beta}_{xy}3d^{2}_{z^2}3d^{\alpha}_{x^2-y^2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d^{2}_{xz}3d^{\beta}_{yz}3d^{\alpha}_{xy}3d^{2}_{z^2}3d^{\alpha}_{x^2-y^2}\right\rangle$$

$$|44,1/2,1/2\rangle = +\frac{1}{\sqrt{2}}\left|3d^{2}_{xz}3d^{\alpha}_{yz}3d^{\beta}_{xy}3d^{\alpha}_{z^2}3d^{2}_{x^2-y^2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d^{2}_{xz}3d^{\beta}_{yz}3d^{\alpha}_{xy}3d^{\alpha}_{z^2}3d^{2}_{x^2-y^2}\right\rangle$$

$$|45,1/2,1/2\rangle = +\frac{1}{\sqrt{2}}\left|3d^{\alpha}_{xz}3d^{2}_{yz}3d^{2}_{xy}3d^{\beta}_{z^2}3d^{\alpha}_{x^2-y^2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d^{\beta}_{xz}3d^{2}_{yz}3d^{2}_{xy}3d^{\alpha}_{z^2}3d^{\alpha}_{x^2-y^2}\right\rangle$$

$$|46,1/2,1/2\rangle = +\frac{1}{\sqrt{2}}\left|3d^{\alpha}_{xz}3d^{2}_{yz}3d^{\beta}_{xy}3d^{2}_{z^2}3d^{\alpha}_{x^2-y^2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d^{\beta}_{xz}3d^{2}_{yz}3d^{\alpha}_{xy}3d^{2}_{z^2}3d^{\alpha}_{x^2-y^2}\right\rangle$$

$$|47,1/2,1/2\rangle = +\frac{1}{\sqrt{2}}\left|3d^{\alpha}_{xz}3d^{2}_{yz}3d^{\beta}_{xy}3d^{\alpha}_{z^2}3d^{2}_{x^2-y^2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d^{\beta}_{xz}3d^{2}_{yz}3d^{\alpha}_{xy}3d^{\alpha}_{z^2}3d^{2}_{x^2-y^2}\right\rangle$$

$$|48,1/2,1/2\rangle = +\frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|49,1/2,1/2\rangle = +\frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|50,1/2,1/2\rangle = +\frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle$$

$$|51,1/2,-1/2\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|52,1/2,-1/2\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|53,1/2,-1/2\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|54,1/2,-1/2\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle$$

$$|55,1/2,-1/2\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|56,1/2,-1/2\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|57,1/2,-1/2\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle$$

$$|58,1/2,-1/2\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|59,1/2,-1/2\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle$$

$$|60,1/2,-1/2\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle$$

$$\begin{aligned}|61,1/2,1/2\rangle =& \sqrt{\frac{2}{3}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\ &- \frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle\end{aligned}$$

$$\begin{aligned}|62,1/2,1/2\rangle =& \sqrt{\frac{2}{3}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\ &- \frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle\end{aligned}$$

$$
\begin{aligned}
|63,1/2,1/2\rangle =& \sqrt{\frac{2}{3}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle \\
&- \frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|64,1/2,1/2\rangle =& \sqrt{\frac{2}{3}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle \\
&- \frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle \\
|65,1/2,1/2\rangle =& \sqrt{\frac{2}{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\
&- \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|66,1/2,1/2\rangle =& \sqrt{\frac{2}{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle \\
&- \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|67,1/2,1/2\rangle =& \sqrt{\frac{2}{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle \\
&- \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle \\
|68,1/2,1/2\rangle =& \sqrt{\frac{2}{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle \\
&- \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|69,1/2,1/2\rangle =& \sqrt{\frac{2}{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle \\
&- \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle \\
|70,1/2,1/2\rangle =& \sqrt{\frac{2}{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle \\
&- \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle \\
|71,1/2,-1/2\rangle =& \frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
&- \sqrt{\frac{2}{3}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle
\end{aligned}
$$

$$
\begin{aligned}
|72,1/2,-1/2\rangle =& \frac{1}{\sqrt{6}}\left|3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta\right\rangle \\
&- \sqrt{\frac{2}{3}}\left|3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha\right\rangle \\
|73,1/2,-1/2\rangle =& \frac{1}{\sqrt{6}}\left|3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\beta\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta\right\rangle \\
&- \sqrt{\frac{2}{3}}\left|3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\alpha\right\rangle \\
|74,1/2,-1/2\rangle =& \frac{1}{\sqrt{6}}\left|3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^2\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^2\right\rangle \\
&- \sqrt{\frac{2}{3}}\left|3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^2\right\rangle \\
|75,1/2,-1/2\rangle =& \frac{1}{\sqrt{6}}\left|3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta\right\rangle \\
&- \sqrt{\frac{2}{3}}\left|3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha\right\rangle \\
|76,1/2,-1/2\rangle =& \frac{1}{\sqrt{6}}\left|3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\beta\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta\right\rangle \\
&- \sqrt{\frac{2}{3}}\left|3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\alpha\right\rangle \\
|77,1/2,-1/2\rangle =& \frac{1}{\sqrt{6}}\left|3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^2\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^2\right\rangle \\
&- \sqrt{\frac{2}{3}}\left|3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^2\right\rangle \\
|78,1/2,-1/2\rangle =& \frac{1}{\sqrt{6}}\left|3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta\right\rangle \\
&- \sqrt{\frac{2}{3}}\left|3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha\right\rangle \\
|79,1/2,-1/2\rangle =& \frac{1}{\sqrt{6}}\left|3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2\right\rangle \\
&- \sqrt{\frac{2}{3}}\left|3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2\right\rangle \\
|80,1/2,-1/2\rangle =& \frac{1}{\sqrt{6}}\left|3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle \\
&- \sqrt{\frac{2}{3}}\left|3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle
\end{aligned}
$$

$$|81,1/2,1/2\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^0\right\rangle$$

$$|82,1/2,1/2\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^\alpha\right\rangle$$

$$|83,1/2,1/2\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^0\right\rangle$$

$$|84,1/2,1/2\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^2\right\rangle$$

$$|85,1/2,1/2\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^\alpha\right\rangle$$

$$|86,1/2,1/2\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^2\right\rangle$$

$$|87,1/2,1/2\rangle = \left|3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^0\right\rangle$$

$$|88,1/2,1/2\rangle = \left|3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^2\right\rangle$$

$$|89,1/2,1/2\rangle = \left|3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$

$$|90,1/2,1/2\rangle = \left|3d_{xz}^2 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha\right\rangle$$

$$|91,1/2,1/2\rangle = \left|3d_{xz}^2 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2\right\rangle$$

$$|92,1/2,1/2\rangle = \left|3d_{xz}^2 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$

$$|93,1/2,1/2\rangle = \left|3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^0\right\rangle$$

$$|94,1/2,1/2\rangle = \left|3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^2\right\rangle$$

$$|95,1/2,1/2\rangle = \left|3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$

$$|96,1/2,1/2\rangle = \left|3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$

$$|97,1/2,1/2\rangle = \left|3d_{xz}^0 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha\right\rangle$$

$$|98,1/2,1/2\rangle = \left|3d_{xz}^0 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2\right\rangle$$

$$|99,1/2,1/2\rangle = \left|3d_{xz}^0 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$

$$|100,1/2,1/2\rangle = \left|3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$

$$|101,1/2,-1/2\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^0\right\rangle$$

$$|102,1/2,-1/2\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^\beta\right\rangle$$

$$|103,1/2,-1/2\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^0\right\rangle$$

$$|104,1/2,-1/2\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^2\right\rangle$$

$$|105,1/2,-1/2\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^\beta\right\rangle$$

$$|106,1/2,-1/2\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^2\right\rangle$$

$$|107,1/2,-1/2\rangle = \left|3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^0\right\rangle$$

$$|108,1/2,-1/2\rangle = \left|3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^2\right\rangle$$

$$|109,1/2,-1/2\rangle = \left|3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$

$$|110,1/2,-1/2\rangle = \left|3d_{xz}^2 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta\right\rangle$$

$$|111,1/2,-1/2\rangle = \left|3d_{xz}^2 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2\right\rangle$$

$$|112,1/2,-1/2\rangle = \left|3d_{xz}^2 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$

$$|113,1/2,-1/2\rangle = \left|3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^0\right\rangle$$

$$|114,1/2,-1/2\rangle = \left|3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^2\right\rangle$$

$$|115,1/2,-1/2\rangle = \left|3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$

$$|116,1/2,-1/2\rangle = \left|3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$

$$|117,1/2,-1/2\rangle = \left|3d_{xz}^0 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta\right\rangle$$

$$|118,1/2,-1/2\rangle = \left|3d_{xz}^0 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2\right\rangle$$

$$|119,1/2,-1/2\rangle = \left|3d_{xz}^0 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$

$$|120,1/2,-1/2\rangle = \left|3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$

The Ground State CSFs are 92 112

$\left\langle 92, \frac{1}{2}, \frac{1}{2} \left| L_x \right| 43, \frac{1}{2}, \frac{1}{2} \right\rangle = -\frac{1}{\sqrt{2}} I$

$\left\langle 92, \frac{1}{2}, \frac{1}{2} \left| L_x \right| 44, \frac{1}{2}, \frac{1}{2} \right\rangle = -\sqrt{\frac{3}{2}} I$
$\left\langle 92, \frac{1}{2}, \frac{1}{2} \left| L_x \right| 63, \frac{1}{2}, \frac{1}{2} \right\rangle = -\sqrt{\frac{3}{2}} I$
$\left\langle 92, \frac{1}{2}, \frac{1}{2} \left| L_x \right| 64, \frac{1}{2}, \frac{1}{2} \right\rangle = -\frac{3}{\sqrt{2}} I$
$\left\langle 92, \frac{1}{2}, \frac{1}{2} \left| L_x \right| 96, \frac{1}{2}, \frac{1}{2} \right\rangle = -I$

Δg_{xx}

$+\zeta \Delta_{92-43}^{-1}$
$+3\zeta \Delta_{92-44}^{-1}$
$+3\zeta \Delta_{92-63}^{-1}$
$+9\zeta \Delta_{92-64}^{-1}$
$+2\zeta \Delta_{92-96}^{-1}$

$\left\langle 92, \frac{1}{2}, \frac{1}{2} \left| L_y \right| 89, \frac{1}{2}, \frac{1}{2} \right\rangle = I$

Δg_{yy}

$+2\zeta \Delta_{92-89}^{-1}$

$\left\langle 92, \frac{1}{2}, \frac{1}{2} \left| L_z \right| 50, \frac{1}{2}, \frac{1}{2} \right\rangle = -\sqrt{2} I$
$\left\langle 92, \frac{1}{2}, \frac{1}{2} \left| L_z \right| 90, \frac{1}{2}, \frac{1}{2} \right\rangle = 2I$

Δg_{zz}

$+4\zeta \Delta_{92-50}^{-1}$
$+8\zeta \Delta_{92-90}^{-1}$